

CytoNorm app TUTORIAL

[cytonorm-App/README.md at main · cipheLab/cytonorm-App](#)

cytonorm-App Public

This app is a Shiny implementation interface for the CytoNorm2 algorithm described in: PMID: 39871681 – Van Gassen et al., CytoNorm2: A method for batch normalization of cytometry data, 2023. The u...

● R Updated 5 minutes ago

(A) Use cases

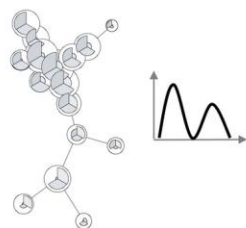
- 1 Normalize without using technical replicates or controls



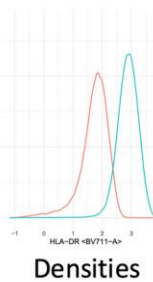
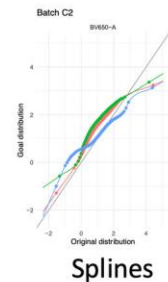
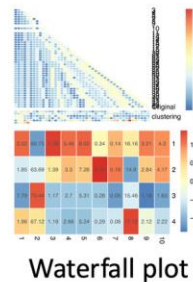
- 2 Normalize towards another model, batch or goal distribution



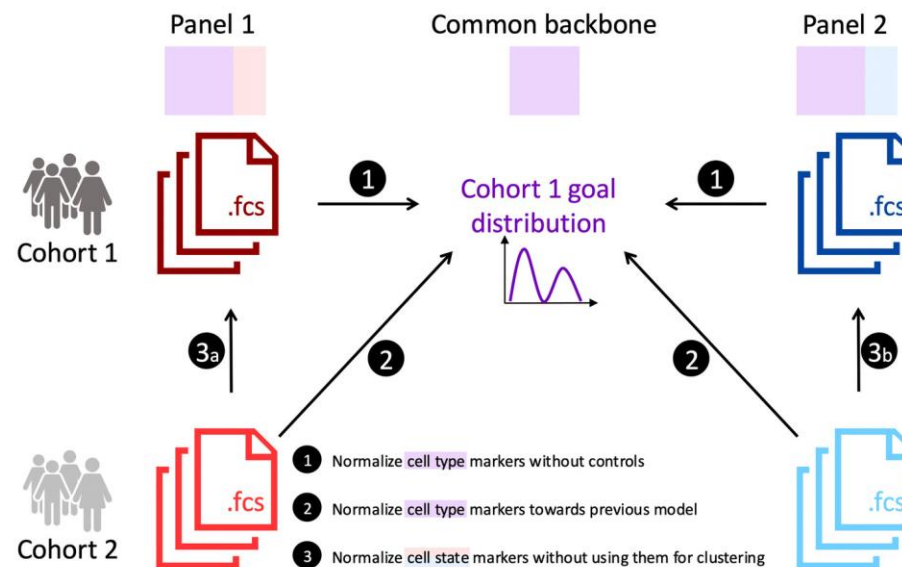
- 3 Normalize markers without including them in the clustering step

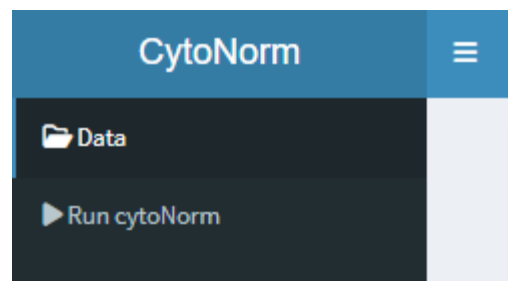


(B) Visualizations



(C) Illustration dataset





STEP 1 : Upload data & choose references

Select batch number

Upload files batch 1

Upload files batch 2

Upload data

Number of batches

2

Batch 1

Upload files for batch 1

Browse...

12-fresh_B6_M1.fcs

Upload complete

Select reference file(s) for this batch!

These reference files will be used by CytoNorm to normalize all other samples in this batch.

Reference files:

12-fresh_B6_M1.fcs

Batch 2

Upload files for batch 2

Browse...

6 files

Upload complete

Select reference file(s) for this batch!

These reference files will be used by CytoNorm to normalize all other samples in this batch.

Reference files:

13-fresh_B6_F1.fcs

What is a Goal Reference?

The goal reference batch is the batch whose distribution will be used as the target for normalization.

All other batches will be normalized towards this batch.

If none is selected, normalization will not use a fixed target batch.

Choose batch as goal reference:

☐ None ☒ Batch 1 ☐ Batch 2

The goal reference batch is the batch whose distribution will be used as the target for normalization.

All other batches will be normalized towards this batch.

If none is selected, normalization will not use a fixed target batch

STEP 2 : Select markers for analysis

Summary table with a column indicating the batch for each file and whether the file is a ref or a sample

Show entries Search:

	Batch	File	Type
1	1	12-fresh_B6_M2-c.fcs	Reference
2	2	8-fresh_TCRbd_M3-c.fcs	Sample
3	2	9-fresh_TCRbd_F2-c.fcs	Sample
4	2	9-fresh_TCRbd_F3-c.fcs	Sample
5	2	10-fresh_PTa_M2-c.fcs	Sample
6	2	10-fresh_PTa_M3-c.fcs	Sample
7	2	11-fresh_PTa_F1-c.fcs	Sample
8	2	12-fresh_B6_M2-c.fcs	Reference
9	2	12-fresh_B6_M3-c.fcs	Sample
10	2	13-fresh_B6_F2-c.fcs	Sample

Showing 1 to 10 of 11 entries Previous 2 Next

Select the markers to be normalized (these will also be the ones that are transformed)

Markers for analysis

H2 (BUV395-A)	Ter119 (BUV496-A)	CD103 (BUV563-A)	CD3e (BUV615-A)	CD11b (BUV661-A)	CD27 (BUV737-A)
TCRb (BUV805-A)	CCR4 (BV421-A)	CD5 (V450-A)	CD2 (BV480-A)	IA/IE (V500-A)	CD24 (BV510-A)
CD117 (BV605-A)	Caspase (BV650-A)	CD161 (NF violet 690-A)	CD71 (BV711-A)	CD40L (BV750-A)	RORgt (BV786-A)
SigF (BB515-A)	Ly6D (FITC-A)	CD4 (Spark Blue 550-A)	CD45 (NovaFluor Blue 585-A)	CD1d (RB613-A)	CCR9 (RB 670-A)
CD69 (BB700-A)	Tbet (RB705-A)	QA2 (RB744-A)	GITR (RB 780-A)	CXCR4 (PE-A)	CD44 (RY586-A)
CD127 (PE-Fire 640-A)	TCRgd (PE-Cy5-A)	icTCRb (RY703-A)	Helios (RY743-A)	CD28 (PE-Cy7-A)	CD8a (NFY 810-A)
CD279 (APC-A)	Eomes (AF660-A)	CD62L (APC-R700-A)	FoxP3 (R718-A)	CD25 (APC-Cy7-A)	

Compensate the data (using the comp matrix in your files)

Transformation: 2 possibilities:
1) Apply the same transformation factor to all channels

Compensation and transformation

☒ Apply compensation if available

Cofactor mode

☒ Same value for all markers

☐ One value per marker

Cofactor value

500

Apply transformation

Transformation: 2 possibilities:
2) Apply a transformation with a different cofactor for each marker

Compensation and transformation

☒ Apply compensation if available

Cofactor mode

☐ Same value for all markers

☒ One value per marker

(Option) Upload EXCEL transformation file

Browse...

No file selected

Apply transformation

Export transfo table

Marker	Fluo	Arg
Time	Time	none
SSC-W	SSC-W	none
SSC-H	SSC-H	none
SSC-A	SSC-A	none
FSC-W	FSC-W	none
FSC-H	FSC-H	none
FSC-A	FSC-A	none
SSC-B-W	SSC-B-W	none
SSC-B-H	SSC-B-H	none
SSC-B-A	SSC-B-A	none
H2	BUV395-A	none
L/D	Zombie UV-A	none
Ter119	BUV496-A	none
CD103	BUV563-A	none
CD3e	BUV615-A	none
CD11b	BUV661-A	none

We test different cluster values for cytonorm. This calculates the coefficient of variation (CV), which measures the variability of markers within clusters between reference batches.

The recommended number of clusters is the one that minimizes the CV, as it corresponds to the best inter-batch stability before normalization.

You can leave the default settings and click the button.

Clustering settings

Markers for clustering

H2 (BUV395-A) Ter119 (BUV496-A) CD103 (BUV563-A) CD3e (BUV615-A) CD11b (BUV661-A) CD27 (BUV737-A) TCRb (BUV805-A)
 CCR4 (BV421-A) CD5 (V450-A) CD2 (BV480-A) IA/IE (V500-A) CD24 (BV510-A) CD19 (BV570-A) CD117 (BV605-A) Caspase (BV650-A)
 CD161 (NF violet 690-A) CD71 (BV711-A) CD40L (BV750-A) RORgt (BV786-A) SigF (BB515-A) Ly6D (FITC-A) CD4 (Spark Blue 550-A)
 CD45 (NovaFluor Blue 585-A) CD1d (RB613-A) CCR9 (RB 670-A) CD69 (BB700-A) Tbet (RB705-A) QA2 (RB744-A) GITR (RB 780-A)
 CXCR4 (PE-A) CD44 (RY586-A) CCR7 (PE-Dazzle594-A) CD127 (PE-Fire 640-A) TCRgd (PE-Cy5-A) icTCRb (RY703-A) Helios (RY743-A)
 CD28 (PE-Cy7-A) CD8a (NFY 810-A) CD279 (APC-A) Eomes (AF660-A) CD62L (APC-R700-A) FoxP3 (R718-A) CD25 (APC-Cy7-A)

FlowSOM cluster selection

Min clusters

5

Max clusters

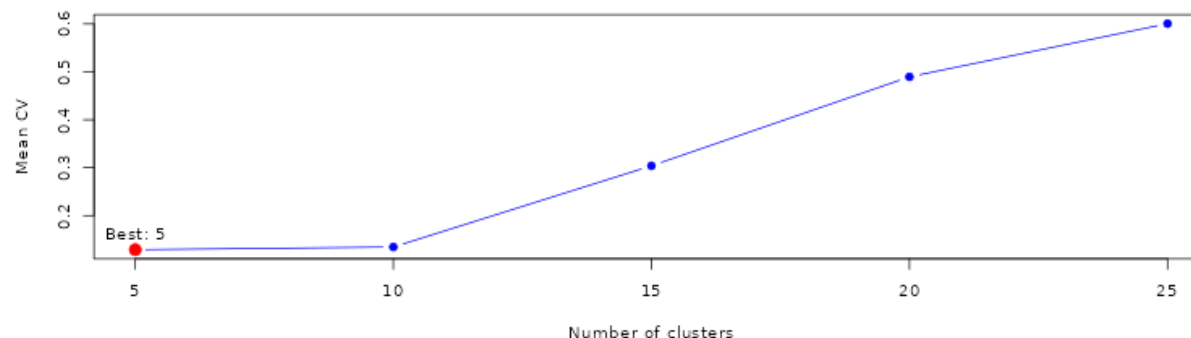
20

Step

5

Test cluster numbers

FlowSOM cluster stability



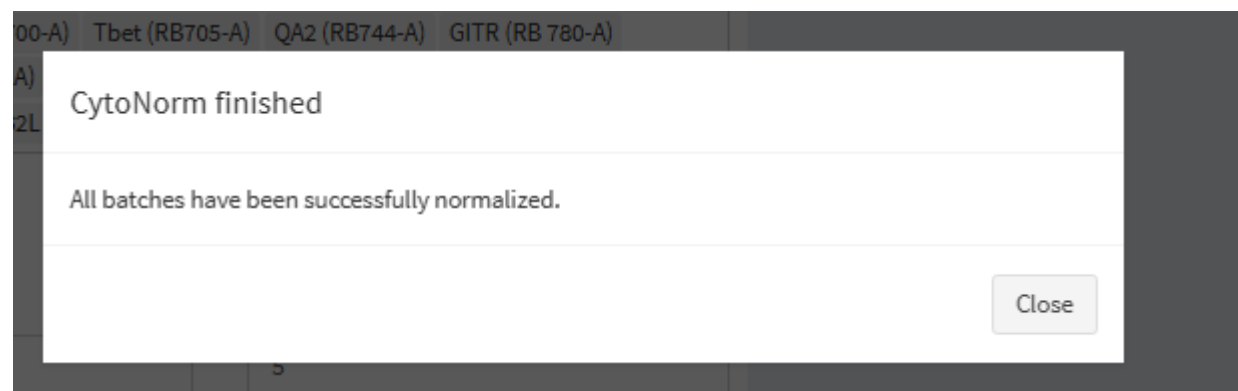
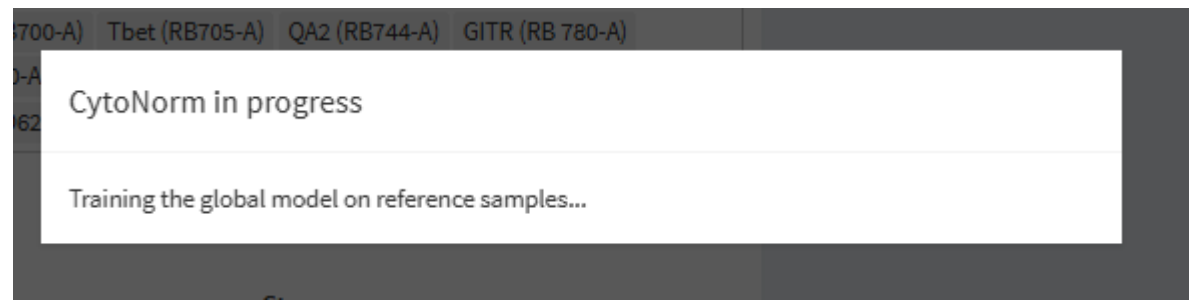
Run Cytonorm

Number of clusters (final run)

Run CytoNorm

Click on this button

& wait until this window appears



Choose untransformed or transformed data and download ZIP

Export normalized files

Download mode:

Decompensated & detransformed|

Compensated & transformed

Decompensated & detransformed

 Download normalized files